

BAAZ JKBOSE



2026 CLASS-11TH

Excretory Products and
Their Elimination

LECTURE-3

Zoology

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Topics To Be Covered

1 Urine Formation

HW Types of Nephron

① Cortical Nephron

→ Short loop of Henle

→ 80%

→ Vasa Recta not well developed

② Juxta Medullary Nephron

→ Long loop of Henle
(Extending into Medulla)

→ 20%

→ Well developed Vasa Recta.

→ Important for Making Concentrated Urine



* Excretory organs in various organism.

① Protonephridia - platyhelminth e.g. Tapeworm.
- Cephalochordata e.g. Amphioxus.
→ It is for Osmoregulation (Water & Salt Balance)

② Nephridia → Annelida e.g. Earthworm

③ Malpighian tubules → Insecta e.g. Cockroach.

④ Antennal Gland / Green Glands → Crabs, prawns.

Aisha
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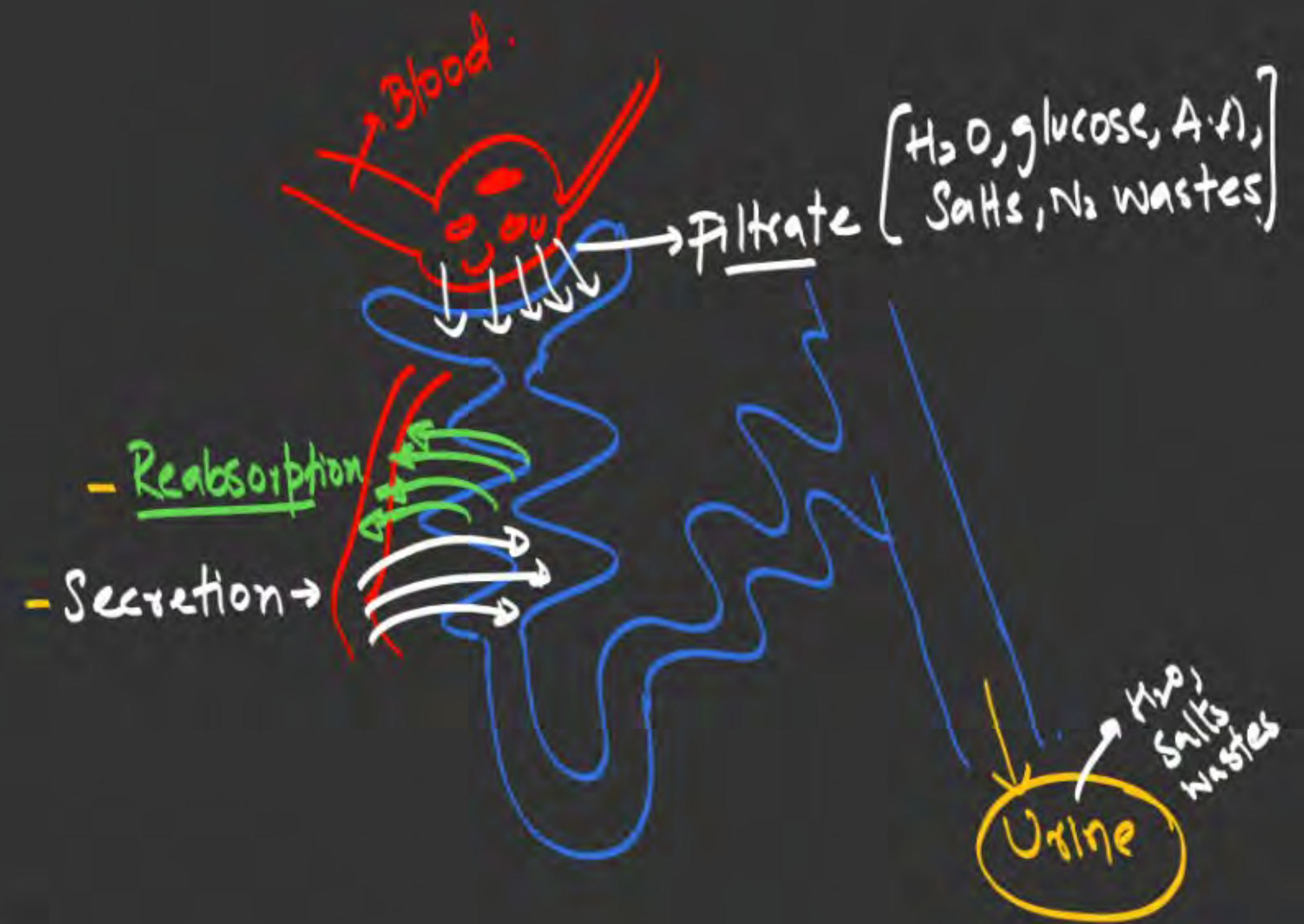
Urine Formation

It involves three steps:

① Ultrafiltration

② Selective Reabsorption

③ Tubular Secretion



① Ultrafiltration

- In this step, some substances leave glomerulus due to filtration pressure and enter into Bowman's capsule.
- These substances form filtrate (primary urine, deproteinized plasma).
- Filtrate contains mainly → H₂O, Ions, Glucose, A.A., N₂ wastes etc

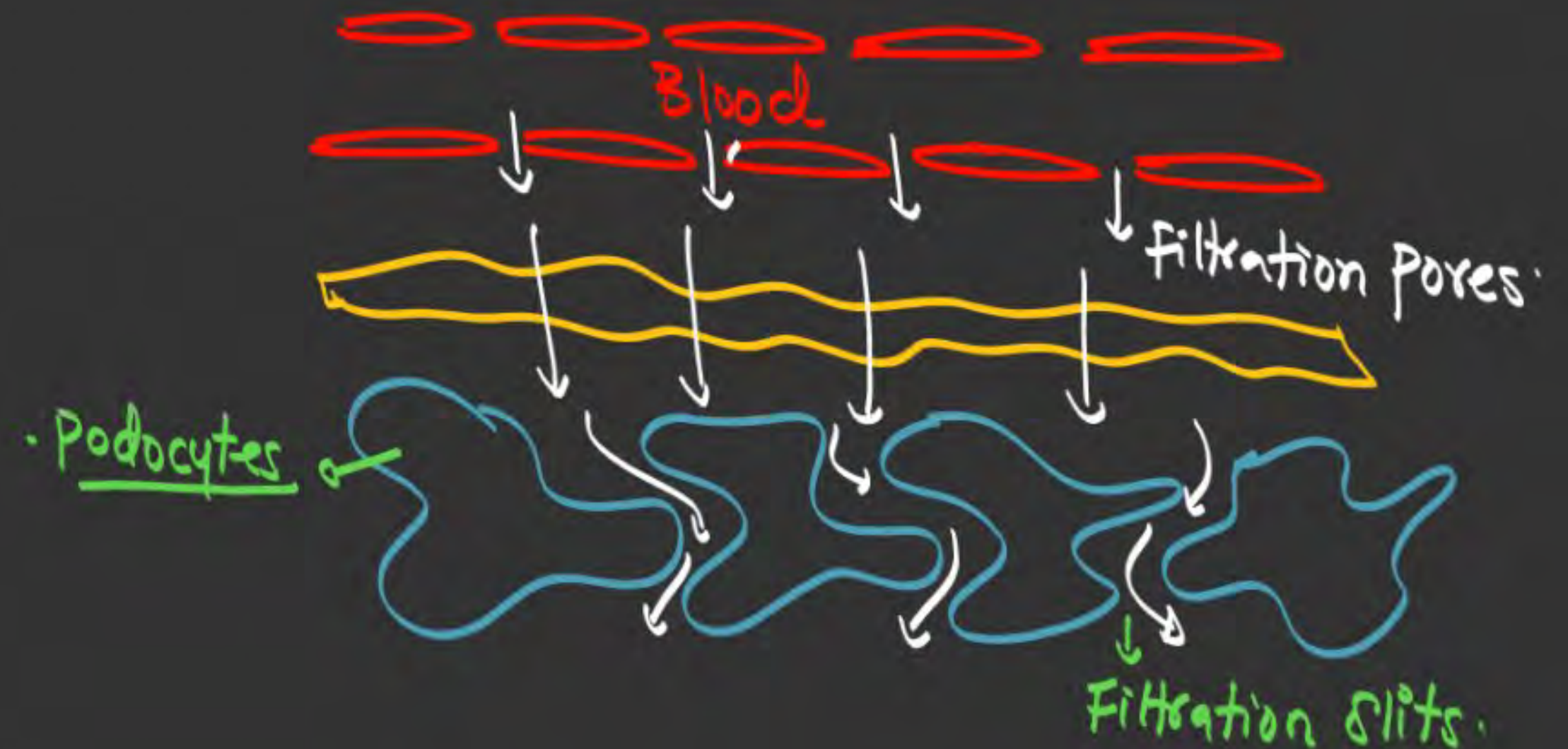
→ Filtration Membrane

It is formed of three layers

- ① Endothelium of glomerulus. (Simple Squamous)
- ② Basement Membrane (Non-cellular)
- ③ Inner (viseral) layer of Bowman's Capsule. (Flat cell - Podocyte)
(Feet like)

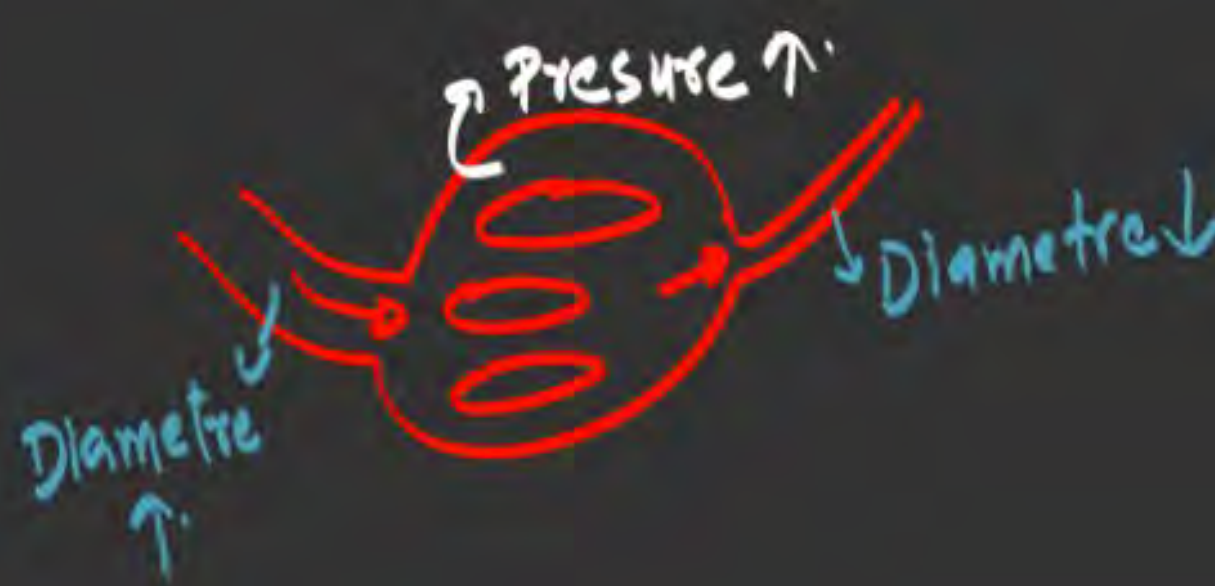


Zoom in.



* Net Filtration Pressure.

↳ one of the Reason of increased pressure in Glomerulus is Difference in Diameters of Afferent & Efferent Arteriole.



$$\hookrightarrow \boxed{NFP = GHP - B COP - CHP}$$

Net Filtration Pressure

Glomerular Hydrostatic Pressure

= 60 mmHg

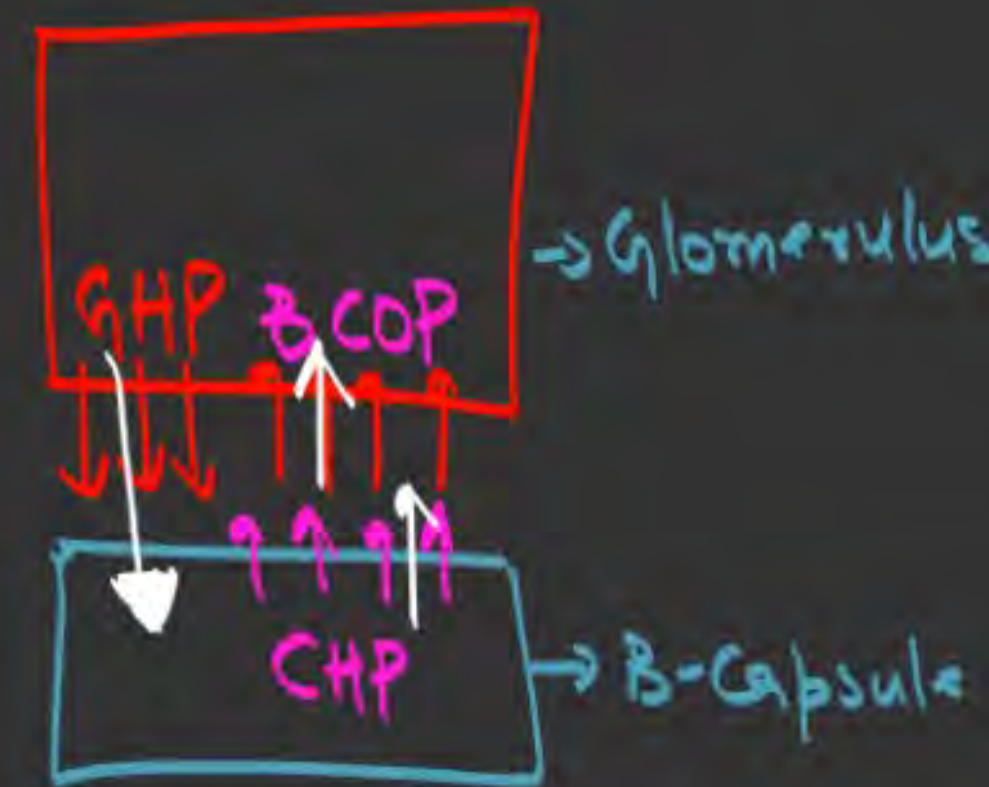
Blood colloidal Osmotic Pressure

[Due to Protein in Blood]
= 30 mmHg

Capsular Hydrostatic Pressure

= 20 mmHg

→ $NFP = +10 \text{ mmHg}$ → Cause filtrate to Move out



* Glomerular Filtration Rate

- Roughly 1100-1200ml of Blood is filtered per Minute by Both kidney.
i.e, $\frac{1}{5^{\text{th}}}$ of Cardiac Output (20%)

- And the Glomerular Filtration Rate = $\frac{125\text{ml}}{\text{Min}}$
(GFR).

$$\begin{aligned}\rightarrow \text{GFR per day} &= 125\text{ml} \times 24 \times 60 \\ &= 180,000\text{ml} \\ &= \underline{180\text{ Litre.}}\end{aligned}$$

→ But Average Urine Formation is 1-1.5 Litre.
→ It Indicates that 99% of Filtrate is Reabsorbed.



⑥+⑦ → Reabsorption & Secretion in Renal tubules

→ PCT

⇒ Reabsorption in PCT

- Max. Reabsorption occurs Here.
- Almost All Nutrients, 70-80% of H_2O & ions
- Filtrate is Isotonic to Blood

⇒ Secretion in PCT

- Secretion of H^+ and NH_3 [Acid Base Balance]
- Secretion of K^+ ions [Ionic Balance]

→ Loop of Henle

→ Reabsorption in Loop of Henle

→ Descending Limb

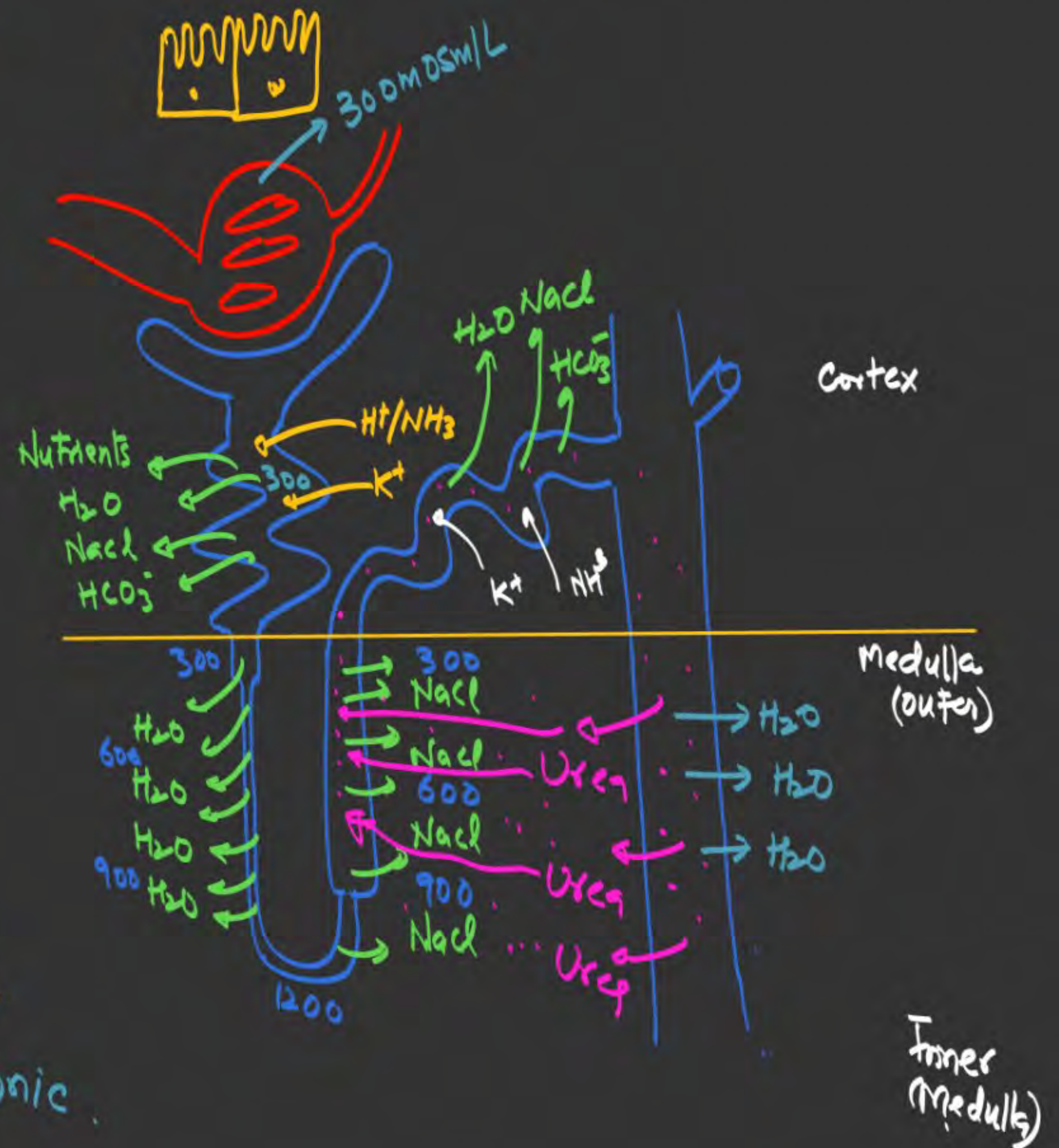
- It is permeable to H_2O & Almost Impermeable to Salts.

→ Filtrate Becomes Hypertonic

→ Ascending Limb

- It is permeable to Salts and Impermeable to H_2O .

→ Filtrate Becomes Hypotonic



⇒ DCT.

- Reabsorption in DCT. (Hypotonic)
 - 'Conditional' Reabsorption of Na^+ & H_2O
 - Na^+ Reabsorption in presence Aldosterone
 - H_2O " " " ADH.
- Secretion in DCT.
 - Secretion of H^+/NH_3 (Acid Base Balance)
 - Secretion of K^+ ion. [Ionic Balance (Na^+/K^+)]

⇒ Reabsorption in Collecting Ducts

- H_2O Reabsorption under ADH.
- Also Collecting Duct are involved in Urea Recirculation.



Osmolarity

→ It is used to measure solvent & solute concentration

→ $\text{Osmolarity} \propto \frac{\text{Solute (salts)}}{\text{Solvent (H}_2\text{O)}}$

→ Osmolarity of Blood
 $= 300 \text{ mOsmoles/Litre}$
 $= \underline{300 \text{ mOsm/L}}$

→ Comparison
 Solutions

Osmolarity ↑ Osmolarity ↓
 ↑ ↓
 Hypertonic Hypotonic

→ Osmolarity same = Isotonic

Thank
You